

Lecture 3

***Before starting our next chapter
(Chapter 22) we will do some
mathematical problems related to
our previous chapter (Chapter 21)***

(For the first 30 min)

Chapter 22
Electric Fields

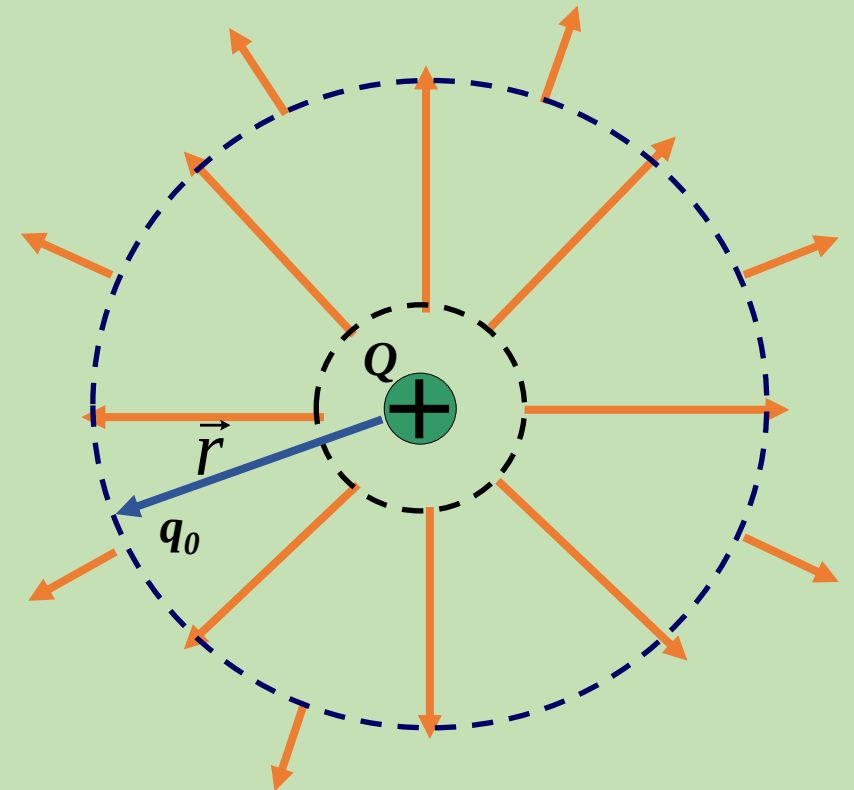
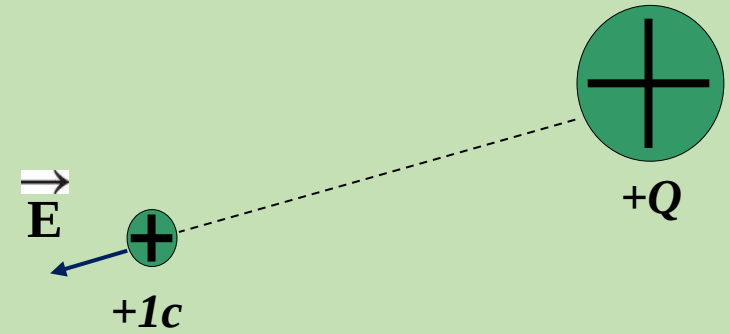
□ Electric Field

The electric field is a force on a unit charge due to a source charge. In other words, the electric field is the force per unit charge.

Suppose, we have a test charge q_0 near a source charge Q and the electrostatic force between them is F . Therefore, the electric field can be written as

$$E = \frac{F}{q_0}$$

The unit of the electric field is NC^{-1}

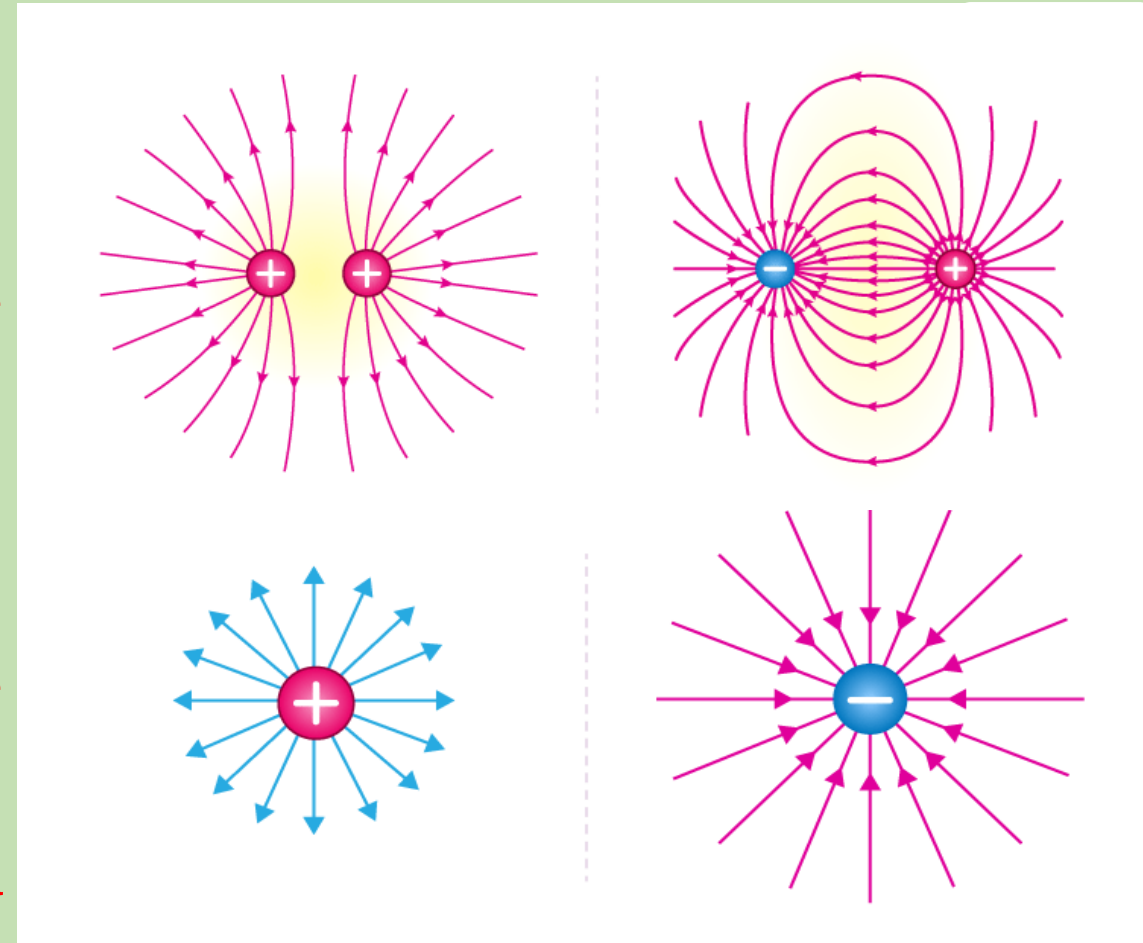


□ Electric Field Lines

When a unit charge is placed near a source charge, it feels force and tries to move to follow a path depending on the force of attraction or repulsion. This path is known as an electric field line.

❖ Properties of Electric Field Lines

- The field lines never intersect each other
- The field lines are perpendicular to the surface of the charges.
- The magnitude of charge and the number of field lines, both are proportional to each other.
- The start point of the field lines is at the positive charge and the end at the negative charge.
- For the field lines to either start or end at infinity, a single charge must be used.



□ The Electric Field due to point charge

To find the electric field due to a charged particle (often called a *point charge*), we place a positive test charge at any point near the particle, at distance r . From Coulomb's law (Eq. 21-4), the force on the test charge due to the particle with charge q is

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r^2} \hat{r}.$$

As previously, the direction of $\vec{F}$ is directly away from the particle if q is positive (because q_0 is positive) and directly toward it if q is negative. From Eq. 22-1, we can now write the electric field set up by the particle (at the location of the test charge) as

$$\vec{E} = \frac{\vec{F}}{q_0} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

Let's think through the directions again. The direction of $\vec{E}$ matches that of the force on the positive test charge: directly away from the point charge if q is positive and directly toward it if q is negative.

So, if given another charged particle, we can immediately determine the directions of the electric field vectors near it by just looking at the sign of the charge q . We can find the magnitude at any given distance r by converting Eq. 22-2 to a magnitude form:

$$E = \frac{1}{4\pi\epsilon_0} \frac{|q|}{r^2}$$

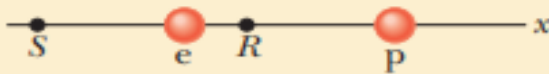
Multiple Electric Fields. The electric field obeys the principle of superposition. Suppose we have n charged particles near a chosen particle called particle 1; then the net electric field on particle 1 is given by the vector sum

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \cdots + \vec{E}_N$$



Checkpoint 1

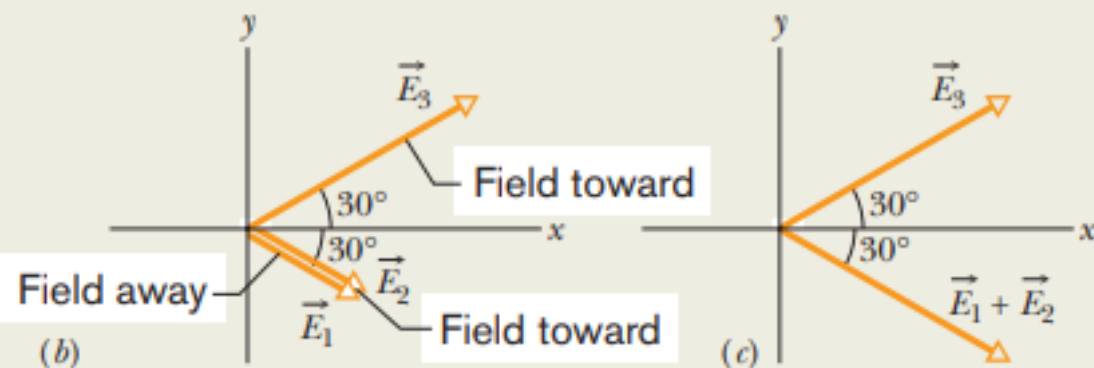
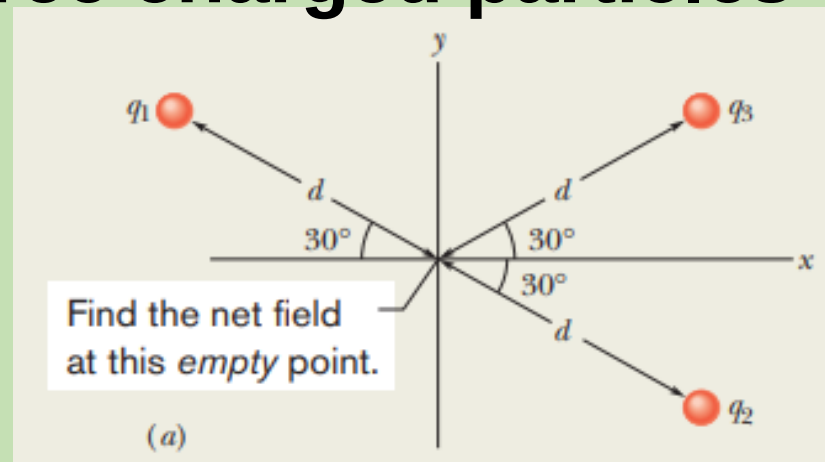
The figure here shows a proton p and an electron e on an x axis. What is the direction of the electric field due to the electron at (a) point S and (b) point R ? What is the direction of the net electric field at (c) point R and (d) point S ?



Net electric field due to three charged particles

Sample Problem 22.01

Figure 22-7a shows three particles with charges $q_1 = +2Q$, $q_2 = -2Q$, and $q_3 = -4Q$, each a distance d from the origin. What net electric field $\vec{E}$ is produced at the origin?



$$E_1 = \frac{1}{4\pi\epsilon_0} \frac{2Q}{d^2}.$$

Similarly, we find the magnitudes of $\vec{E}_2$ and $\vec{E}_3$ to be

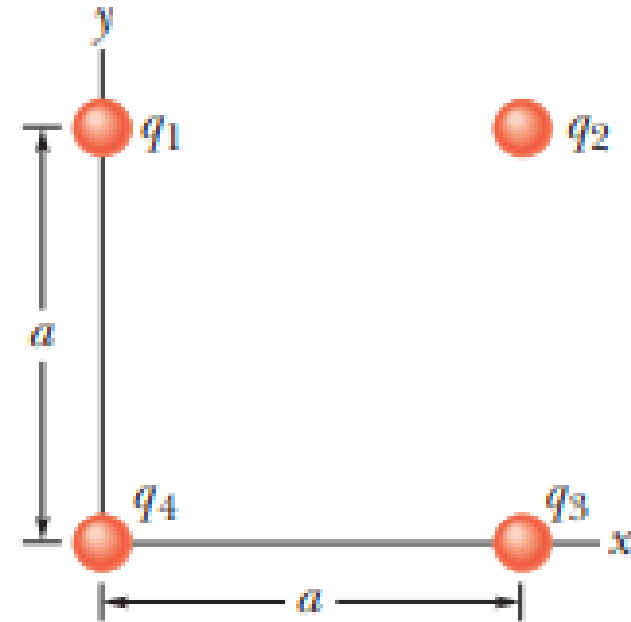
$$E_2 = \frac{1}{4\pi\epsilon_0} \frac{2Q}{d^2} \quad \text{and} \quad E_3 = \frac{1}{4\pi\epsilon_0} \frac{4Q}{d^2}.$$

$$\begin{aligned} E_1 + E_2 &= \frac{1}{4\pi\epsilon_0} \frac{2Q}{d^2} + \frac{1}{4\pi\epsilon_0} \frac{2Q}{d^2} \\ &= \frac{1}{4\pi\epsilon_0} \frac{4Q}{d^2}, \end{aligned}$$

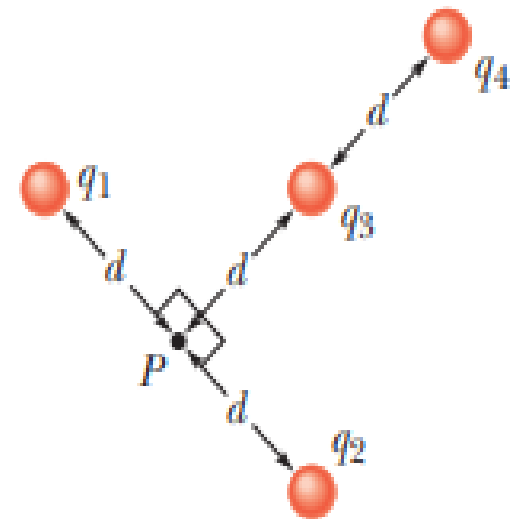
$$\begin{aligned} E &= 2E_{3x} = 2E_3 \cos 30^\circ \\ &= (2) \frac{1}{4\pi\epsilon_0} \frac{4Q}{d^2} (0.866) = \frac{6.93Q}{4\pi\epsilon_0 d^2}. \quad (\text{Answer}) \end{aligned}$$

Unsolved Problems related to Electric Fields

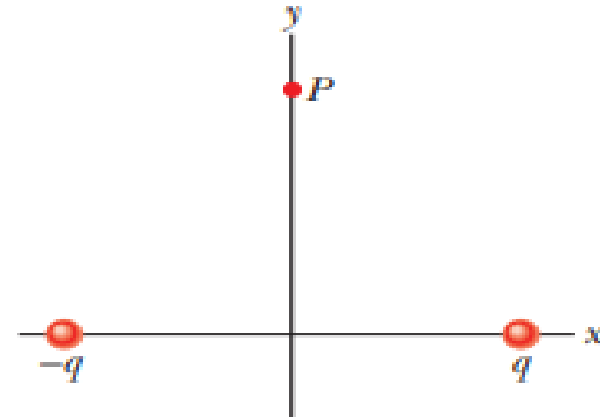
••7 SSM ILW WWW In Fig. 22-35, the four particles form a square of edge length $a = 5.00$ cm and have charges $q_1 = +10.0$ nC, $q_2 = -20.0$ nC, $q_3 = +20.0$ nC, and $q_4 = -10.0$ nC. In unit-vector notation, what net electric field do the particles produce at the square's center?



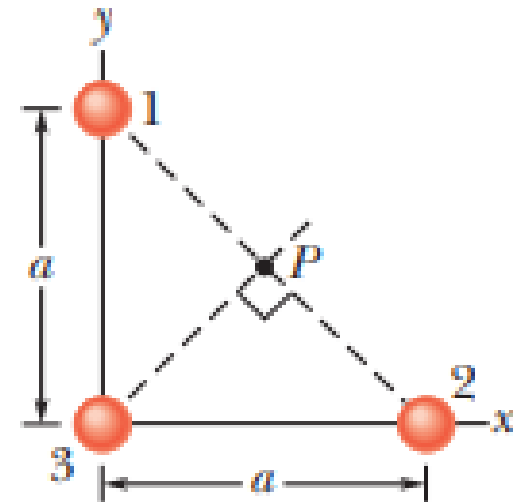
••8 GO In Fig. 22-36, the four particles are fixed in place and have charges $q_1 = q_2 = +5e$, $q_3 = +3e$, and $q_4 = -12e$. Distance $d = 5.0$ μm . What is the magnitude of the net electric field at point P due to the particles?



••9  Figure 22-37 shows two charged particles on an x axis: $-q = -3.20 \times 10^{-19} \text{ C}$ at $x = -3.00 \text{ m}$ and $q = 3.20 \times 10^{-19} \text{ C}$ at $x = +3.00 \text{ m}$. What are the (a) magnitude and (b) direction (relative to the positive direction of the x axis) of the net electric field produced at point P at $y = 4.00 \text{ m}$?



••15 In Fig. 22-42, the three particles are fixed in place and have charges $q_1 = q_2 = +e$ and $q_3 = +2e$. Distance $a = 6.00 \mu\text{m}$. What are the (a) magnitude and (b) direction of the net electric field at point P due to the particles?



Thank You